

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Physics Semester I Examination 2024-25 (NEP)

DEPARTMENT OF PHYSICS**Mid-Semester Examination****Paper Code: PHYMJ11 / PHYMN11 — Mechanics and General Properties of Matter**

Time: 1:00 Hour

Maximum Marks: 20

*Instruction: Attempt all questions. Each question carries 5 marks.***Question 1**

[5]

What are the effects of fictitious/apparent forces appearing due to Earth's rotation?

Question 2

[5]

Explain inertial and non-inertial frames of reference with examples.

OR

The radius of Earth is $R = 6.37 \times 10^8$ cm, its mean density is $\rho = 5.5$ g/cm³, and the gravitational constant is $G = 6.66 \times 10^{-8}$ CGS units. Calculate the Earth's surface gravitational potential.

Question 3

[5]

Establish a relation between Young's modulus (Y) and Bulk modulus (K) of an elastic material. Show that:

$$Y = 3K(1 - 2\sigma)$$

where σ is Poisson's ratio.**Question 4**

[5]

Show that there can be four points of suspension in a compound pendulum about which the time periods of oscillation are equal.

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